

11 **(short)** A coaxial cable is shown in Fig. 6 that has a core conducting wire with a radius of R_1 , and a coaxial cylinder with inner and outer radii of R_2 and R_3 , respectively. The gap is filled with an insulator material that has relative permeability μ_r . Current, I , is flowing bottom up in the core wire and flowing top down in the coaxial cylinder, as indicated by Fig. 6. Draw the magnetic flux density profile in the whole coaxial cable as a function of radius (r), i.e., $0 \leq r \leq R_3$. [10]

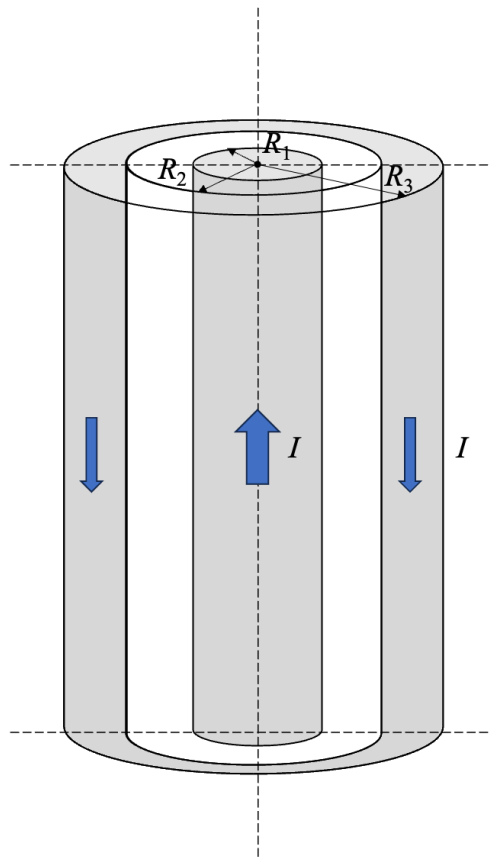


Fig. 6